

Due on March 2nd, 2026 (im terrified)

The main point is below. If you're overwhelmed just read this part below please. The data to look at is at the bottom of the doc.

Phase 1: Ice Hockey Performance Predictions

You're the analytics staff for the World Hockey League. Your group is preparing for the World Hockey League (WHL) Tournament, and the WHL commissioner has a set of tasks for you. You will explore the numbers from the current WHL season to make predictions of what teams will likely succeed in the tournament. In this competition, your group will step into the role of sports data scientists to predict tournament matchups and dive deeper to uncover features underlying team performance.

Your challenge is to formulate your responses based only on the provided, fictional World Hockey League data set. While these data are meant to reflect real-world ice hockey, they do not represent any actual teams or leagues.

Your analytics staff's challenge is to:

1a: Create a power ranking of WHL teams based on a season of performance data and predict the winners of tournament match-ups.

1b: Rank the top 10 WHL teams on offensive line quality disparity.

1c: Communicate the impact of offensive team line quality disparity using a data visualization.

1d: Tell your story and summarize your methodology. Craft a concise but comprehensive explanation of your approach.

You and your team have been hard at work analyzing and using the data to come up with the best answers. Now it's time to share your predictions!

Keys to winning (read this if you want to win)!!!!!!!

Success in this competition is about showcasing your skills in three key areas:

Accuracy: How effective are your matchup predictions and rankings of team depth? How clear was your visualization of the impacts of team depth on game outcomes?

Methodology: How well-thought are your quantitative approaches to creating matchup predictions and offensive depth rankings? How clear and accurate is your visualization? Can you explain your approach clearly and justify your choices with sound reasoning?

Communication: Do your findings tell a compelling story, both visually and in your presentation?

This is more than a numbers game—it's a chance to turn data into strategy, strategy into predictions, and predictions into high-level sports analytics grounded in real-world ice hockey dynamics.

crunch the numbers to provide an outlook for the WHL tournament, using the provided in-season data.

These next ones go more in depth about phase 1

Phase 1a: Team Performance Analysis

Before diving into Phase 1a, first create a league table outlining teams' overall standings for the season following the online educational module using your method of choice (R, Python, Google Sheets, GenAI/LLM). The league table will not be submitted on its own but is a necessary first step to creating team power rankings and predicting game outcomes for your Phase 1a submission.

Create team power rankings: Based on season data, rank the 32 teams using the underlying team performance. But this isn't just about win-loss records. Your rankings should reflect the overall strength and quality of the teams.

Predict game outcomes and win probabilities: For the first round of the World Hockey League tournament, predict the win probability for the home team for 16 matchups.

Submissions: Power ranking of all 32 teams and winning probabilities of home teams in 16 matchups submitted by Student Team Leader via SurveyMonkey Apply

Phase 1b: Line Performance Analysis

Quantify team offensive line quality disparity

Hockey teams use multiple offensive lines and defensive pairings. The first offensive line is often more productive than the secondary lines — meaning they tend to generate more scoring opportunities. Your task is to quantify how large that disparity is for each team.

First, for each offensive line on each team, form an offensive performance measure based on expected goals (xG). Consider accounting for differences in time on ice so teams or lines that play more don't automatically appear better, and defensive matchups since tougher opponents can affect performance.

Second, for each team, compare performance of the first line to the secondary line by calculating a ratio of the first line's offensive performance measure to that of the secondary line. This ratio represents the team's offensive line quality disparity. Which teams have the largest offensive line quality disparity? Rank teams from largest offensive line quality disparity to smallest.

Submission: A list of the Top 10 teams with the largest offensive line quality disparity ratios (ranked 1 through 10) submitted by Student Team Leader via SurveyMonkey Apply.

Phase 1c: Data Visualization

Communicate the impact of offensive team line quality disparity

The WHL commissioner wants to know if teams with more evenly-matched offensive lines are more likely to succeed. Your task is to provide insight into how offensive line quality disparity (calculated in 1b) may relate to team strength (calculated in 1a) in a data visualization.

Your visualization should:

- Clearly communicate your main message or finding.

- Use labeled axes, titles/subtitles, captions, and legends to make your graphic understandable without additional explanation.
 - Reflect sound data practices as described in the educational modules.
- Submission:* single PNG file upload (max file size 5MB) submitted by Student Team Leader via SurveyMonkey Apply.

Phase 1d: Tell Your Story - Summarize Your Methodology

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story. Your task is to craft a concise but comprehensive explanation of your approach—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that another analytics group could replicate your methodology, showcasing the rigor and transparency of your work.

Your summary should cover these key points:

(1) Process:

- How did you clean or transform the raw data before analysis? Please describe it in around 50 words.
- Did you create any additional variables? Please describe in around 25 words.

(2) Tools and Techniques:

- What software tools did you use? [Select all that apply]
- How did you use the selected software, and for what purposes? Please describe in around 50 words.
- What statistical methods did you employ? Please describe in around 100 words.

(3) Your Predictions:

- 1a: How did you determine power rankings and matchup probabilities? Please describe in around 50 words.
- 1b: How did you approach summarizing offensive line quality disparity? Please describe in around 50 words.
- 1c: What data visualization choices did you make to clarify the numbers and emphasize takeaways? Please describe in around 50 words.

(4) Your Insights:

- How did you assess your model performance (i.e., how well any models explain the data)?
- Did you use Generative AI tools (ChatGPT, Gemini, etc.)? If so, explain how—describe each stage where AI tools came into the process and how your group approached using them. Please describe in around 50 words.

Data links (May God, Buddha, Brahma, Zeus, Allah bless us and get us through this)

- [Game data set \(whl_2025\)](#), primary data set, with game level line summary data
- [WHSDSC_2026_DataDictionary](#) for all fields in the Game box scores file
- [WHSDSC 2026 Glossary](#): Glossary of terms
- [WHSDSC_Rnd1_matchups](#): Round 1 game matchups to predict in Phase 1a
- Another way to get to above links: Google Drive [WHSDSC_2026_CompetitionPackage](#) folder
- [Upenn hockey explanation for competition](#) [Video Library](#)